

Number, units and estimation - Set 1 - Solutions

Specification M1.1, M1.2, M2.1-M2.14.

Answer key

Question	Answer	Topic
1	D	Number, units and estimation
2	C	Number, units and estimation
3	D	Number, units and estimation
4	F	Number, units and estimation
5	F	Number, units and estimation
6	D	Number, units and estimation
7	B	Number, units and estimation
8	E	Number, units and estimation
9	D	Number, units and estimation
10	B	Number, units and estimation
11	F	Number, units and estimation
12	F	Number, units and estimation

Worked solutions

Question 1 NUM-01 · Number, units and estimation · spec M2.3

Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

Answer: D

Take the lower power of each common prime: $2^2 \times 3^2 = 36$.

Question 2 NUM-03 · Number, units and estimation · spec M2.9

Express the recurring decimal $0.\dot{2}\dot{7}$ as a fraction in its lowest terms.

Answer: C

$x = 0.2727\dots$, $100x = 27.2727\dots$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$.

Question 3 NUM-04 · Number, units and estimation · spec M2.13

A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

Answer: D

Upper bounds 4.65 and 3.5: $4.65 \times 3.5 = 16.275$.

Question 4 NUM-05 · Number, units and estimation · spec M2.8

Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.

Answer: F

$$\frac{24 \times 10^2}{6 \times 10^{-3}} = 4 \times 10^5.$$

Question 5 NUM-06 · Number, units and estimation · spec M2.5

A code consists of two letters from A, B, C, D, E followed by two digits from 1 to 9. Letters may be repeated but the two digits must be different. How many codes are possible?

Answer: F

$$5 \times 5 \times 9 \times 8 = 1800.$$

Question 6 NUM-07 · Number, units and estimation · spec M1.2

A car travels at 90 km/h. What is this speed in metres per second?

Answer: D

$$90\,000 \text{ m in } 3600 \text{ s: } 90000/3600 = 25 \text{ m/s.}$$

Question 7 NUM-08 · Number, units and estimation · spec M2.14

Estimate the value of $\frac{4.92 \times \sqrt{99.3}}{0.198}$ by rounding each number to 1 significant figure.

Answer: B

$$\frac{5 \times 10}{0.2} = 250.$$

Question 8 NUM-10 · Number, units and estimation · spec M2.13

A number x is truncated (not rounded) to 2 decimal places and the result is 3.14. Which inequality describes the possible values of x ?

Answer: E

Truncation removes the later digits, so x is at least 3.14 and less than 3.15.

Question 9 NUM-11 · Number, units and estimation · spec M2.2

Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.

Answer: D

$$\frac{7}{3} - \frac{7}{4} = \frac{28-21}{12} = \frac{7}{12}; \frac{7}{12} \times \frac{6}{7} = \frac{1}{2}.$$

Question 10 NUM-13 · Number, units and estimation · spec M2.7

Which of the following is equal to $\frac{8^4 \times 2^{-5}}{4^3}$?

Answer: B

$$\frac{2^{12} \cdot 2^{-5}}{2^6} = 2^1 = 2.$$

Question 11 NUM-14 · Number, units and estimation · spec M2.12

Simplify $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$.

Answer: F

$$\sqrt{48} = 4\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}. \text{ So } \frac{4\sqrt{3} \times 3\sqrt{3}}{2\sqrt{3}} = \frac{36}{2\sqrt{3}} = \frac{18}{\sqrt{3}} = 6\sqrt{3}.$$

Question 12 NUM-15 · Number, units and estimation · spec M2.5

How many three-digit positive integers have digits that are all different and all odd?

Answer: F

Odd digits 1, 3, 5, 7, 9: $5 \times 4 \times 3 = 60$.